



LUMENCORE
LUMAARC SEAL OF APPROVAL
MARKET RESEARCH CAPABILITY STATEMENT
Project Argos Agentic AI Proof of Concept
Sources Sought Notice ONC-ARGOS-SSN-2026-OS351107

DRAFT - HUMAN REVIEW AND ACTION-TIME FACTS REQUIRED

Required cover field	Current response
Company name	LumenCore
Responding legal entity	Pending action-time fact
Seal	LumaArc seal of approval
UEI / DUNS if applicable	Pending action-time fact
Company address	Pending action-time fact
Authorized point of contact	Pending action-time fact
Telephone / email	Pending action-time fact
Small-business designation(s)	Pending action-time fact
Public reviewer surface	https://lumen-core.ai/
Public evidence repository	https://github.com/robertashworth1986-debug/lumen-core-public
Response deadline	July 30, 2026 at 5:00 PM Eastern

Market research response only. No proprietary, classified, confidential, CUI, patient, or sensitive information is included.

1. Executive Fit and Recommended Role

LumenCore offers Project Argos a bounded evidence-assurance and deterministic-validation workstream: authorized-source custody, rule traceability, hash-locked evidence cases, human decision gates, adverse-result retention, and reviewer handoff. These are component capabilities supported by the public LumenCore repository and public reviewer surface.

For Project Argos, those patterns align most directly with requirements traceability, public-source ingestion provenance, deterministic validator orchestration, evidence-case generation, corrective-action retest records, and production handoff documentation. LumenCore is responding at component level and does not claim presently qualified full-scope health IT prime readiness. A credible team still requires named FHIR R4/CHPL and ONC Certification Program expertise plus an HHS ATO/FIPS 199/security lead.

Recommended acquisition position: bounded evidence-assurance workstream participant in a named team. Full-prime readiness is not claimed.

2. Understanding of the Requirement

- Argos is a governed monitoring and evidence system, not a general-purpose chatbot. Standards-based and legal pass/fail decisions should remain deterministic wherever possible, with AI limited to bounded support such as summarization, classification, reconciliation, and drafting.
- The evidence chain begins with authoritative public sources and must preserve raw artifacts, request context, timestamps, response metadata, normalized records, lineage decisions, validation outputs, and hashes.
- Suspected non-conformities require human review, rule-level traceability, deduplication, plain-language narratives, controlled corrective-action support, and before/after retesting.
- Real-world endpoint observation must be safe and unauthenticated where appropriate, must avoid patient data and protected health information absent separate written authorization, and must operate within approved rate, scope, security, and hosting boundaries.
- Production readiness includes Government-aligned deployment, operations and rollback runbooks, cost and staffing visibility, and an HHS authorization path; a proof-of-concept result alone is not an Authority to Operate.

3. Task-by-Task Capability and Teaming Matrix

The matrix distinguishes current evidence-backed contribution from work that must be led or accepted by qualified health IT, regulatory, cybersecurity, prime, and Government personnel.

Task	Requirement	Proposed position	Bounded contribution
1	Initiation, requirements, and regulatory traceability	Core contribution with partner review	Build a versioned requirements-to-test-to-evidence matrix; define evidence identifiers, severity fields, human checkpoints, decision rights, schedules, and change-control receipts. A health-regulatory lead must approve mappings to 45 CFR and ONC program practice.
2	Monthly administrative reporting	Core contribution	Produce source-backed status reports with task progress, blockers, expected completion dates, evidence links, decisions, and unresolved gates.
3	Public-source discovery and ingestion	Core contribution with FHIR partner	Implement source inventory, raw-response custody, timestamps, request metadata, hashes, lineage, precedence rules, data dictionaries, and reproducible normalized projections. Partner validates CHPL, Lantern, NPPES, and FHIR domain mappings.

Task	Requirement	Proposed position	Bounded contribution
4	Agentic architecture and governance	Core contribution with security partner	Separate deterministic validators from AI support; version policies, prompts, rules, retries, rate limits, fallbacks, error handling, and audit logs; require human authorization for compliance determinations.
5	Publication and FHIR R4 conformance testing	Partner-led	LumenCore can wrap validator outputs in traceable evidence cases and regression receipts. A qualified FHIR/ONC partner must own standards interpretation, test design, and conformance acceptance.
6	Real-world endpoint observation	Partner-led with evidence support	Contribute safe-observation logging, scope controls, drift records, hashes, and replayable comparison artifacts. Partner owns health-endpoint semantics, Government-approved access boundaries, and PHI avoidance controls.

Task	Requirement	Proposed position	Bounded contribution
7	Evidence case generation and triage	Core contribution	Assemble raw and normalized artifacts, validator output, request/response logs, timestamps, hashes, severity, rule traceability, issue narratives, deduplication keys, and machine/human-readable exports.
8	Corrective-action workflow support	Core contribution with regulatory approval	Provide controlled draft, review, authorization, retest, and before/after evidence workflows. ONC/ONC-ACB discretion and regulatory interpretation stay with authorized personnel.
9	HHS Authority to Operate	Security-partner-led	LumenCore can supply component inventories, reproducibility records, change receipts, evidence links, and control-test artifacts. The team requires an experienced HHS ATO/FIPS 199/SSP lead and authorized assessment support.

Task	Requirement	Proposed position	Bounded contribution
10	Production-ready PPC release package	Core contribution with prime	Package code, dependency locks, manifests, runbooks, rollback procedures, limitations, test coverage, operating assumptions, costs, staffing needs, and reviewer receipts. Hosting and production acceptance remain prime/Government decisions.
11	Public-artifact assessment and FY 2027 strategy	Core contribution with domain partner	Extend the same traceability and evidence-case pattern to approved public documentation checks, prototype scoring only under locked criteria, and retain uncertainty and adverse findings in expansion recommendations.

4. Proposed Evidence-Assurance Workstream

LumenCore proposes a fail-closed sequence in which missing authority, provenance, validation, or human disposition remains visible rather than becoming a promoted compliance claim.

1. Authorize sources and scope: Record approved source, purpose, collection boundary, cadence, rate limits, and prohibited data before collection.
2. Preserve raw observations: Store source payloads and request context with UTC timestamps, immutable identifiers, and SHA-256 manifests.
3. Normalize with lineage: Create common records while retaining source precedence, conflicts, and reversible links to raw artifacts.
4. Run deterministic checks: Apply versioned rules and validators; retain rule IDs, inputs, outputs, failures, and environment details.

5. Use AI only inside policy: Permit bounded summaries, clustering, reconciliation, and draft narratives; block autonomous compliance determinations.
6. Assemble an evidence case: Package artifacts, logs, hashes, timestamps, severity, traceability, narrative, and unresolved questions in human- and machine-readable forms.
7. Require human disposition: Route suspected issues to authorized reviewers and record decisions, rationale, corrective-action drafts, and retest outcomes.

5. Component Evidence and Similar-Scope Status

The following public records support the engineering patterns offered here. They are not presented as healthcare, agency, field, or economic validation.

Evidence record	What the public artifact supports	What it does not support
Public reviewer capsule	31 of 31 declared assertions and 3 of 3 suites reproduced in the packaged clean-run workflow, with dependency and source-state checks.	First-party bounded reproducibility only; not external validation, agency certification, field performance, or health IT past performance.
Custody and validation controls	Versioned manifests, SHA-256 receipts, schema checks, duplicate-action locks, and fail-closed gate records support an inspectable evidence workflow.	Control artifacts do not establish health-domain correctness, production authorization, contract performance, or agency acceptance.
Adverse-result retention	Public records preserve failed promotion gates, negative findings, and unresolved authorities instead of converting them into favorable claims.	Transparent failure handling is an engineering pattern, not proof of FHIR conformance, regulatory interpretation, or field performance.

The notice requests experience of similar scope and complexity. The matrix below distinguishes LumenCore component evidence from qualifications that must come from an authorized team; it does not substitute adjacent technical work for federal health prior performance.

Capability area	Present support	Acquisition implication
Evidence custody, traceability, and deterministic validation	Component pattern supported by public code, tests, receipts, and reviewer artifacts.	LumenCore can be evaluated for a bounded evidence-assurance workstream.

Capability area	Present support	Acquisition implication
FHIR R4, CHPL/Lantern/NPPES, and ONC Certification Program delivery	No direct LumenCore prior-performance reference is claimed.	Named and authorized health IT/FHIR partner evidence is required.
HHS ATO, FIPS 199, SSP/control implementation, and 3PAO coordination	No direct LumenCore HHS authorization reference is claimed.	Named and authorized federal cybersecurity lead and independent assessment path are required.
Full-scope federal health program integration	No full-prime readiness or comparable federal health delivery is claimed.	A qualified small-business prime or integrated team must own program delivery and provide relevant references.

The public evidence index states the present boundary directly: current strengths are artifact custody, deterministic replay, fail-closed claim governance, reviewer handoff packaging, and bounded pilot design. Independent scientific validation, field-validated savings, agency endorsement, certified safety, audited revenue, and customer adoption require separately identified external records.

6. Delivery and Security Approach

- Operate in a Government-approved environment with explicit authorization boundaries, least privilege, authenticated administration, dependency locking, secrets separation, immutable audit records, and rollback procedures.
- Treat HHS ATO as a managed authorization program: FIPS 199 categorization, boundary definition, SSP/control implementation, evidence collection, assessment coordination, POA&M handling, and Authorizing Official review.
- Do not collect, request, store, or use patient data or PHI unless separately authorized in writing and supported by approved architecture and controls.
- Keep AI components advisory and inspectable. Every model, prompt, policy, validator, retry, fallback, and rule version must be traceable to the evidence case it influenced.

Current boundary: the public repository is not an HHS-authorized production system and does not establish an ATO, FHIR certification, 3PAO assessment, or federal health deployment. Those are explicit team and acquisition gates.

7. Teaming Structure and Missing Qualifications

Workstream	Accountability	Current gate
LumenCore evidence-assurance workstream	Source custody, deterministic validation orchestration, traceability, evidence cases, hashes, claim boundaries, reviewer packages, retest receipts.	Public repository evidence supports bounded reproducibility patterns.
Health IT/FHIR and ONC program lead	CHPL/Lantern/NPPES semantics, FHIR R4 Endpoint/Organization/Bundle testing, ONC regulatory mapping, corrective-action content review.	Pending named and authorized partner
Federal cybersecurity/ATO lead	HHS authorization boundary, FIPS 199, SSP, control implementation, assessment coordination, POA&M and authorization package.	Pending named and authorized partner
Prime/program integration lead	Contract performance, staffing, Government coordination, hosting, delivery acceptance, health-domain prior performance, and integrated schedule.	Pending named and authorized partner

No partner name, credential, contract vehicle, clearance, prior-performance reference, or legal designation will be included without authorization and documentary support at action time.

8. Illustrative Mobilization Plan

The schedule below is an acquisition-planning outline, not a binding offer. It must be reconciled with Government scope, hosting, staffing, review cadence, security boundary, and the SOW's authorization milestones.

Period	Illustrative outcome
0-30 days	Confirm team, scope, source authorization, regulatory ownership, hosting assumptions, traceability schema, security boundary, and pilot cohort.

Period	Illustrative outcome
31-90 days	Stand up bounded ingestion, lineage, deterministic validator interfaces, evidence-case schema, human work queue, baseline test corpus, and audit logs.
91-180 days	Expand approved endpoint observation, drift tracking, issue clustering, corrective-action draft/retest workflow, security documentation, and operational runbooks.
181 days and beyond	Complete Government-directed hardening, assessment evidence, production handoff, PPC evaluation, cost/staffing analysis, and FY 2027 strategy.

9. Questions and Requested Next Step

1. What proof-of-concept duration, target cohort size, and Government hosting environment should respondents assume for acquisition planning?
2. Which organization owns final interpretation of 45 CFR and ONC Certification Program test outcomes, and what review service levels are expected?
3. Will HHS provide an approved source list, rate-limit policy, synthetic test corpus, and expected evidence-case export schema?
4. Is HHS seeking one full-scope prime, or would it value clearly bounded small-business teaming responses for evidence assurance, FHIR validation, and ATO workstreams?
5. What existing HHS authorization boundary, reusable controls, continuous-monitoring services, or 3PAO arrangements may be available to the PPC?

Requested next step: include LumenCore in market-research or teaming discussions where a qualified prime needs an inspectable evidence-assurance workstream for source custody, deterministic validators, rule traceability, human-gated issue disposition, and reproducible handoff.

This response is for market research only and is not an offer, proposal, certification, or representation that every draft SOW task is presently covered.